



Impacts of generative AI on user contributions: evidence from a coding Q&A platform

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Abstract

This study investigates the short-term impact of generative AI, exemplified by the introduction of ChatGPT, on user contributions in a coding Q&A platform. We find that the introduction of ChatGPT led to a reduction in the number of high-quality answers provided by users, particularly among highly engaged contributors, despite an overall increase in answers. We identify two key mechanisms: (1) increased perceived question sophistication despite no actual change in content and (2) reduced motivation of loyal users in providing answers in the face of AI-generated alternatives. The findings suggest that while generative AI can facilitate value creation on user-generated content (UGC) platforms, it also poses challenges in retaining core contributors and managing content quality. The paper contributes to the literature on the impact of AI adoption on platforms and suggests practical implications for UGC platform management, such as the need for AI content disclosure measures to retain engaged users.

Keywords Generative AI · User contribution · UGC platform · Content policy

1 Introduction

Large language models (LLMs) are fundamentally transforming content creation and knowledge exchange among users in online communities. These advanced generative intelligences have shown remarkable potential in quickly providing high-quality answers to questions that require in-depth understanding and specific domain knowledge (Liévin et al., 2023; OpenAI, 2023). Their superiority over conventional search engines and unengaged online communities poses a direct challenge to user-generated content (UGC) platforms that rely on Q&A functionalities.

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On one hand, generative AI can facilitate value-creating interactions on these UGC platforms by empowering content creation and addressing low-response problems. For creators, such AI assistance can lower their costs to compose complete answers with more detailed information, while shortening their creation and publishing cycle (Epstein et al., 2023). At the same time, this efficiency improvement might increase the satisfaction of those who initiate the discussion and seek answers, further promoting positive interactions between users. For platforms, AI-generated content can serve as an innovative solution to cold start problems concerning lack of discussions for early-stage projects (Hou et al., 2022).

On the other hand, it may discourage user participation through several mechanisms: reducing the social rewards for creators due to reduced audience reach (Zhang & Zhu, 2011) and increasing the costs of producing distinctive content to gain visibility (Lysyakov & Viswanathan, 2023). As a result, this could crowd out voluntary contributions of high-quality content. Furthermore, while generative AI challenges incumbent users' position in content production, this could also create difficulties for platforms' content moderation, potentially flooding them with misinformation (Bubeck et al., 2023). Consequently, the spread of misinformation could lead to users' distrust and disrupt future content production and consumption. Recent studies find concerning evidence on LLM hallucination, where false statements are erroneously assumed to be true by AI, resulting in the generation of falsehoods (Bang et al., 2023).

With these, there is limited understanding of the impact of generative AI on UGC platforms, especially those centered around knowledge sharing via Q&A discussions. To fill this gap, we are particularly interested in answering the following two specific questions: (1) *Does the release of LLM-based generative AI affect content creation on UGC platforms powered by Q&A features?* (2) *If so, what are the primary mechanisms driving this impact?* By answering these questions, we hope to uncover the role of generative AI in knowledge creation and explore strategies to better position AI tools and AI-generated content in the value-creation process on UGC platforms.

To address these research questions, we collected user post data from a large Q&A community. Leveraging the introduction of ChatGPT as a natural experiment, we examined the causal effects of generative AI on the answers posted by users on the platform. We first observed a large-scale deletion of suspicious AI-generated answers, indicating users' active adoption of ChatGPT. Our Difference-in-Differences (DID) analyses then demonstrated a reduction in the number of high-quality answers provided by the same users, partially due to decreased demand for answers. Beyond this shift in demand, our research highlights another demand-driven aspect: the increase in perceived question sophistication—despite no actual change in content—further reduces the number of high-quality answers. Finally, we find that the decline in answer contributions is primarily driven by highly engaged contributors who regularly provide solutions, indicating that the AI tool mainly demotivated core users.

Our research contributes to the literature on the impact of AI adoption on online platforms. Prior studies have demonstrated the positive impacts of AI on user interactions in online marketplaces (Bell et al., 2023; Brynjolfsson et al., 2019; Liu et al., 2023). However, obstacles remain in applying AI to consumer-focused services due to AI aversion (Castelo et al., 2023; Longoni et al., 2019; Puntoni et al., 2021; Schmitt, 2020). Anthropomorphic strategies have been proposed but fall short in facilitating

seamless human-AI interactions, as AI is often perceived as lacking human-like capabilities (Bergner et al., 2023; Hermann, 2022). Recent advancements in generative AI may reshape this perception and transform the human-AI interaction landscape (Brynjolfsson et al., 2023; Noy & Zhang, 2023). Recent work on online communities shows that generative AI can largely displace these knowledge communities to meet crowds' information acquisition demand, but this negative effect can be mitigated by strong community culture (Burtch et al., 2024). Complementing this line of research, our study sheds light on the impact of generative AI on loyal platform users who actively engage in knowledge exchange and content moderation. The observed decline in their contributing effort suggests a reserved attitude towards these technologies among core community contributors (Lysyakov & Viswanathan, 2023).

Second, our study complements prior studies on managing UGC platforms. Extant literature shows that motivating and sustaining user content creation is crucial for the survival of a UGC platform (Ahn et al., 2016; Khurana et al., 2019). Prior studies have suggested several solutions, such as monetary rewards and reputation systems, to incentivize content contribution (Burtch et al., 2022; Liu & Feng, 2021). However, limited empirical research exists on the effectiveness of content regulation policies, particularly regarding openness and copyrights (Li et al., 2021; Wlömert et al., 2023). Our study examines the effect of a policy banning AI-generated content on user behavior, showing its limited effectiveness. Finally, our study offers practical implications for platforms. Understanding the dynamics of content creation, especially in the context of generative AI, is essential for informed policy development. Our study suggests that platforms should consider implementing AI content disclosure measures rather than directly removing susceptible AI-generated content. This approach may be more effective in retaining the core UGC contributors.

2 Data and empirical model

2.1 Data and variables

We conducted an empirical study on Stack Overflow, a widely recognized Q&A platform.¹ We chose Stack Overflow for its extensive popularity among programmers, with a substantial user base of proficient and passionate programmers who actively contribute numerous answers daily to address programming queries. Given the potential impact of generative AI, we anticipate that its influence will be particularly pronounced on Q&A platforms where users voluntarily provide answers. Therefore, platforms like Stack Overflow, which primarily focus on coding queries, offer an ideal setting for observing and analyzing the significant effects of generative AI on UGC contributions.

We procured data through the Stack Exchange Data Explorer query tool. Our dataset includes 679,662 answers and 773,613 questions posted during the months of November and December for the years 2021 and 2022. For each post, we collected various information such as the full textual content, engagement metrics (votes), deletion status, and details about the users who posted and edited the content. Additionally,

¹ <https://stackoverflow.com>

information such as titles, attached tags, and the number of answers at the time of data collection were included for questions. These contributions came from 188,853 unique users for answers and 309,346 unique users for questions. On average, questions on Stack Overflow received 0.93 answers during this period.

To examine the impact of generative AI, we leverage the unexpected release of ChatGPT on November 30, 2022, as a natural experiment. ChatGPT, OpenAI's chatbot powered by LLM, represents a significant advancement in the content generation capability of AI. Its release has ignited a competitive race among tech giants to develop comparable LLMs and enhance their products with this newfound capability. Developers are also actively exploring creative applications and sharing their experiences on social media and programming communities. Given the significance of ChatGPT as a milestone in familiarizing the public with AI, its release serves as an ideal event for investigating our research questions.

Taking the introduction of ChatGPT as an exogenous shock, we constructed a symmetric 9-day time window from November 26 to December 4 to collect all posted answers and questions. A key empirical challenge is to establish counterfactuals for estimating the average treatment effect on the treated (ATT) on online community users' contributions. While we cannot identify the average treatment effect on actual ChatGPT adopters due to the lack of information on individuals' variational adoption timings (Burtch et al., 2018), we can exploit the simultaneous availability of ChatGPT to all users to estimate its overall impact on community engagement. We achieve this by employing a variant of the DID approach, with the posts from the same timeframe in the preceding year as a control group (Brodeur et al., 2021; Goldberg et al., 2024). This approach allows us to capture the broader community impact, rather than focusing solely on early adopters, thus estimating the average effect of exposure to the new generative AI environment, including non-adopters (Xu et al., 2024). Furthermore, the short time window design enables us to isolate the impact of ChatGPT's release from the temporary usage ban that occurred on December 5.² In a later section, we will extend the timeframe to explore the long-term effects of ChatGPT.

We measure users' contribution activities in two common dimensions: contribution intensity and quality (Huang et al., 2019). Contribution intensity is gauged by calculating each user's daily creation volume, while contribution quality is assessed through viewers' responses to answers or questions, using vote-based scores as proxies. Specifically, the score of a post is defined as the number of upvotes minus the number of downvotes from viewers. In our data, undeleted answers, on average, received a score of 0.89, while deleted answers received an average score of -0.55. This suggests that undeleted answers generally exhibit higher quality, a trend also observed for questions.

In addition to post-related statistics, we extract textual features from the post content³ to gain additional insights into post quality. It is important to note that these specific textual measures are unavailable for deleted posts due to platform regulations. Following common practice in UGC literature (Ghose et al., 2012), we employ text analytics tools to compute metrics such as text length, readability metrics, and

² <https://meta.stackoverflow.com/questions/421831/temporary-policy-generative-ai-e-g-chatgpt-is-banned>

³ We exclude code blocks from the extracted text to ensure accuracy.

information entropy. These objective linguistic measures provide a comprehensive view of quality by capturing the effort involved in user contributions aside from peer reviews. Other potential user-related factors including user characteristics and historical activities are collected as well. A summary of these variables and their statistics is provided in Table A1 of our Supplementary Material).

2.2 Empirical model

We employ a DID analysis to assess users' responses to the introduction of generative AI at the user-day level. The introduction of ChatGPT on November 30, 2022, is treated as a natural experiment, with the post-treatment period occurring after November 30 and the pre-treatment period occurring before. Our treated group comprises answers created in 2022, while the control group consists of those created in the preceding year.

To build our user sample, we focus on users who consistently contributed to the platform in both the pre-treatment and post-treatment periods of 2021 and 2022. We identified 546 users who posted answers throughout all four distinct periods. This user panel serves as a representative sample to study the impact on users' answering activities. Our primary model can be formulated as follows:

$$Y_{ijt} = \beta_0 + \beta_1 \text{TreatYear}_j \times \text{After}_t + u_i + \tau_t + Z_{ijt} + \varepsilon_{ijt}, \quad (1)$$

where Y_{ijt} denotes the quantity or quality measures of an answer, as defined in Table A1. These dependent variables are measured based on the daily creations of user i on day t in year j during the study window. The variable TreatYear_j is a dummy variable indicating whether an answer is created in 2022. After_t denotes whether an answer is created after the cutoff date (i.e., November 30). The terms u_i and τ_t represent user and day fixed effects, respectively. We also include several time-varying controls related to a user's experience and recent activity, such as UserTenure (the number of days since a user joined the platform) and $\text{UserPastContribution}$ (the total number of answers posted by a user in the past 10 days).

3 Impact on answer contributions

3.1 Model-free evidence

To provide model-free evidence of the effect of ChatGPT, we analyze the temporal trends in answer volumes in 2022 and compare them to the corresponding period in 2021.⁴ Based on the platform's content moderation policy, we categorize answer postings into those that were removed and those that were not. We then present the daily total number of answers and the ratio of undeleted answers in Fig. 1.

⁴ We have aligned the plots by the same weekdays to facilitate a clearer comparison of changes in total answer quantity and the ratio of undeleted answers between 2021 and 2022.

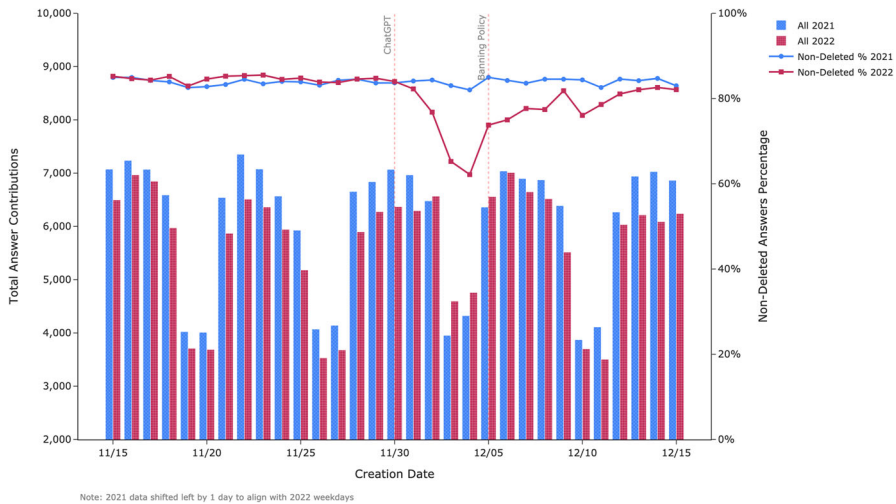


Fig. 1 Temporal trend of daily answer postings and the ratio of undeleted answers

Figure 1 shows the temporal patterns in answer volumes on the platform. In the weeks leading up to the introduction of ChatGPT, daily answers (including both deleted and non-deleted) were generally lower than in the previous year. However, there was a significant, albeit temporary, increase in overall answer postings relative to the previous year just a few days after ChatGPT's availability. This increase subsequently disappeared, and the contributions in 2022 generally remained lower than in 2021. The sharp drop in the ratio of non-deleted answers after ChatGPT's availability suggests that the increase was primarily driven by the proliferation of deleted answers. This observation aligns with our expectation that AI tools like ChatGPT can lower barriers to answer composition, resulting in more AI-generated answers at the expense of user-generated ones. In response, the platform implemented a policy to regulate AI-generated content on December 5, 2022, leading to a surge in deleted answers on that day and the subsequent dates. An examination of the creation dates of these deleted answers revealed that nearly 95% were created after the release of ChatGPT, indicating a large-scale influx of AI-assisted answers. However, when focusing on undeleted answer volumes, it appears that the daily number of undeleted answers has actually decreased.⁵ We will investigate this in more detail in the following sections. Finally, the trends between 2021 and 2022 exhibit consistent parallel patterns until the disruption caused by the event on November 30, providing support for our following year-on-year DID comparison.⁶

⁵ The decline in undeleted answer postings is evident from a simple calculation, as the increase in overall quantity cannot offset the sharp decrease in the undeleted ratio.

⁶ We plot the time series in other proximate years in Fig. A1 as well. Among these time series, the parallel pattern holds reasonably well between 2021 and 2022, suggesting a relatively stable content creation market in those 2 years.

3.2 Main effects

We report the main results regarding answer contributions. Column 1 of Table 1 presents the regression results, with the number of undeleted answers as the dependent variable. The coefficient of particular interest, $TreatYear \times After$, is significantly negative, suggesting a notable 11.2% decrease in the number of undeleted answers following the introduction of ChatGPT, as compared with the same period in 2021. This negative effect shrinks to 4.7% after we consider the daily demand for questions in column 2, indicating that decreased demand for answers could partly explain fewer undeleted answers.⁷ Furthermore, our results show that users who have contributed more answers, particularly those with more accepted answers in the past, tend to contribute more answers currently. This suggests a positive relationship between past user engagement, especially in terms of providing valuable answers, and current answer contributions.

Regarding the quality of answers, we begin by examining the effect on the helpfulness votes of answers. Column 3 in the table reveals a 0.122 decrease in quality score following the release of ChatGPT. This decline, if modest, signals a significant shift in the effort users devote to crafting solutions to programming inquiries. Further examination through textual analysis, as detailed in columns 4–6, suggests that there is no notable change in key textual characteristics of answers such as length, readability, and information richness in the short run.

The observed reduction in both answer quantity and quality suggests that users are contributing fewer high-quality answers after ChatGPT became available. This trend may result from several factors. First, users might be making less effort to answer questions on the platform, possibly because they are less motivated to provide original answers given the influx of AI-generated responses. Second, the decline in answers can be an outcome of shifting demand for programming solutions. Users probably find chatbots' advantages in solving their puzzles in comparison to forum discussions. This substitution can then lead to a change in the content and presentation of questions on Stack Overflow. That is, questions might become more intricate and challenging to answer because simple questions are easily addressed using ChatGPT. We test these possible mechanisms in the later sections.

4 Possible mechanisms of driving fewer high-quality answer contributions

4.1 Change in the presentation and content of questions

We investigate whether the decreasing supply of high-quality answers was influenced by a potential shift toward more demanding questions. With answer seekers now using ChatGPT for basic coding instructions, if they start asking more challenging questions, it may pose challenges for answer providers to meet the increased demand

⁷ We also attach the results on using a logarithmized supply-to-demand ratio to examine the decline in answer contribution. Consistent results from Table D1 provide more robustness to our main findings.

Table 1 Results of DID estimations for answers

	Quantity		Quality	Quality (text)		
	(1) Undeleted	(2) Undeleted	(3) Score	(4) Length	(5) Readability	(6) Entropy
TreatYear $\times$ After	−0.112*** (0.019)	−0.047** (0.020)	−0.122* (0.073)	−0.002 (0.044)	0.097 (0.123)	−0.035 (0.026)
TreatYear	0.025 (0.016)	0.030* (0.016)	−0.290*** (0.092)	−0.014 (0.035)	−0.121 (0.099)	0.010 (0.021)
UserTenure	−0.025 (0.020)	−0.025 (0.020)	0.173 (0.116)	−0.022 (0.046)	0.191 (0.173)	0.014 (0.030)
UserPastContribution	0.011*** (0.002)	0.011*** (0.002)	−0.005 (0.003)	0.003 (0.002)	−0.007 (0.004)	0.001 (0.001)
UserAcceptedRate	0.033 (0.027)	0.032 (0.027)	−0.170 (0.133)	0.015 (0.068)	0.218 (0.201)	0.010 (0.041)
QuestionVolume		0.413*** (0.047)	−0.074 (0.248)	−0.155* (0.080)	−0.392 (0.241)	−0.086* (0.048)
User FE	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9822	9822	5347	5347	3697	5347
Adjusted R^2	0.303	0.310	0.154	0.481	0.323	0.525
F	22.890	33.446	8.337	1.379	1.594	0.767

Note: The table reports the estimation results of answers at the user and day level. *Readability* is measured by the SMOG index, which refers to how many years of education the average person needs to have to understand a text. *UserPastContribution* and *UserAcceptedRate* refer to the number of answers posted and the proportion of answers that were selected as accepted answers in the last 10 days, respectively. Standard errors are clustered at the user level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

for sophisticated answers. This could result in a temporary decline in the quantity or quality of provided answers, particularly if contributors are unable or unwilling to meet the heightened expectations.

To examine this hypothesis, we analyze the impact on the presentation and content of questions. However, due to the limited number of users posing questions consistently throughout the four distinct periods, constructing a user-day level panel dataset is not feasible. Consequently, we rely on post-level data for our analysis of questions (see our model specification in Supplementary Material). It is worth noting that we control for question categories to account for inherent differences across programming domains, such as complexity and generalizability.

Our results provide evidence supporting the hypothesis that the release of ChatGPT led to a shift in question presentation. As shown in Table 2, there was no noticeable change in question scores (column 1). However, we observed a 0.031 rise in the text entropy of questions (column 4). Since text entropy measures the diversity of vocabulary and structure usage, the increase indicates an elevation in perceived question sophistication from users' perspectives. This observation motivates us to further examine potential shifts in question content.

Table 2 Results of DID estimations for questions

	Quality	Quality (text)			Text similarity	
	(1) Score	(2) Length	(3) Readability	(4) Entropy	(5) TF-IDF	(6) Doc2Vec
TreatYear × After	0.041 (0.065)	0.028 (0.020)	0.027 (0.101)	0.031** (0.013)	−0.001 (0.003)	−0.002 (0.004)
TreatYear	−0.510*** (0.056)	−0.025 (0.019)	0.075 (0.073)	−0.021 (0.013)	−0.001 (0.003)	−0.012* (0.006)
UserTenure	0.041 (0.036)	0.033*** (0.012)	0.081 (0.049)	0.017** (0.007)	0.001 (0.001)	−0.001 (0.004)
UserPastQuestions	−0.016* (0.009)	0.005 (0.006)	0.020 (0.016)	0.003 (0.003)	−0.000 (0.001)	−0.002 (0.001)
CategoryPastQuestions	0.008 (0.209)	−0.003 (0.072)	−0.353 (0.346)	0.003 (0.049)	−0.010 (0.014)	−0.008 (0.023)
User FE	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,307	19,307	14,380	19,307	19,307	19,307
Adjusted R^2	0.080	0.410	0.278	0.439	0.637	0.205
F	38.135	3.184	2.701	2.403	0.681	4.851

Note: The table reports the DID estimation results of questions at post level. *Readability* is measured by the SMOG index, which refers to how many years of education the average person needs to have to understand a text. *TF-IDF* and *Doc2Vec* refer to the min-max normalized cosine similarity that is established on TF-IDF and Doc2Vec PV-DM (distributed memory model of paragraph vectors), respectively. Standard errors are clustered at the category level

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

To examine potential changes in question content, we assess question novelty as a more appropriate proxy. While the previous analyses indicate changes in question presentation, the actual content of the questions may not have been significantly altered. It is possible that ChatGPT merely assists users in expressing the same content in a novel manner. To evaluate question content, we compute question novelty scores by measuring the average similarity of each question to preceding ones. We calculate the average cosine distance between each question and a reference pool consisting of questions posted in the 30 days before our study period. Following existing research on UGC (Burtch et al., 2022), we employ two widely used NLP embedding techniques, term frequency-inverse document frequency (TF-IDF) and Doc2Vec, to convert texts into numerical vectors. TF-IDF incorporates relative word importance, while Doc2Vec leverages contextual information and semantic meanings for text embeddings.⁸ The last two columns of Table 2 indicate no change in the actual content of the questions, suggesting that the perceived increase in question sophistication is likely unrelated to a fundamental change in question content but rather reflects the presentation of questions. Consequently, the decrease in high-quality answer contributions cannot be

⁸ Detailed model descriptions and training process can be found in Supplementary Material.

directly attributed to differences in the questions themselves. Instead, the increase in perceived question sophistication, as indicated by the rise in text entropy, may have led viewers to perceive coding instruction requests as more complex. This perception could then contribute to a decrease in the willingness to provide answers.

4.2 Demotivation effect of ChatGPT

Next, we investigate whether the increasing number of AI-generated answers demotivates certain users from providing answers. Existing loyal users may feel challenged to differentiate their answers from AI-generated ones and struggle to stand out, potentially leading to a loss of interest in contributing. This effect may be more pronounced among users with extrinsic motivations, such as career concerns, as the surge in AI-generated answers could discourage their participation and even prompt them to leave the community. Additionally, loyal users who voluntarily provide answers may perceive a diminished social value in their contributions and experience a reduced sense of belonging to the community when they witness a multitude of easily generated AI answers. On the other hand, new or novice users may be more inclined to embrace ChatGPT and use it to showcase their ability to provide answers.

To assess the demotivating effect, we assessed how changes in answer contributions were influenced by users' prior engagement levels. As shown in Table 3, a notable decline of 8.1% in daily contributions from heavy users—who were previously more active—drove the decrease. In contrast, light users experienced an insignificant impact. In addition, the average quality score for heavy users dropped by 0.149, whereas this metric remains largely unchanged for light users.⁹ Furthermore, the positive coefficient for the smog readability index and the negative coefficient for text entropy, though not significant, implies a possible decline in the effort users are willing to invest in their contributions. This shift could stem from the perception that the value and recognition of contributing high-quality, information-rich answers are diminished when AI-generated content proliferates, potentially reducing the perceived social rewards of participating in the community.

The heterogeneous effects on answer contributions between heavy and light users remain consistent across several robustness checks. For instance, using longer pre-periods or considering overall activity levels still indicates that the decrease in answer contributions is primarily driven by heavy users (Table C1, columns 1–3). This gap is consistent or even more pronounced when we consider other user characteristics that relate to motivation. While senior users cut down their contribution more than junior users, users who highly value their reputation seem to be the most discouraged from contributing answers when they notice a rise in AI-generated responses (Table C1, columns 4 and 5). Additionally, the fact that the difference between senior and junior users is smallest provides further evidence that it is not seniority, but the importance placed on reputation, that primarily drives the demotivation effect. Regarding the quality of answers, the coefficient of interest is not consistently significant (Table C2).

⁹ For a detailed examination of the differences between the two groups, please refer to Figs. A3 and A4, which depict histograms comparing users' daily contributions and average scores before and after the shock.

Table 3 Heterogeneity analysis of answers

	Quantity (1) Undeleted	Quality (2) Score	Quality (text)		
			(3) Length	(4) Readability	(5) Entropy
Panel A: High					
TreatYear × After	−0.081*** (0.029)	−0.149* (0.084)	−0.024 (0.050)	0.122 (0.142)	−0.040 (0.030)
TreatYear	−0.018 (0.031)	−0.362*** (0.127)	−0.026 (0.047)	0.076 (0.143)	−0.005 (0.028)
UserTenure	0.064 (0.069)	0.648*** (0.154)	−0.033 (0.119)	−0.445 (0.539)	0.017 (0.078)
UserAcceptedRate	0.090 (0.065)	−0.246 (0.205)	0.047 (0.124)	−0.258 (0.329)	0.036 (0.072)
QuestionVolume	0.475*** (0.064)	−0.135 (0.310)	−0.174* (0.094)	−0.419 (0.283)	−0.095* (0.055)
User FE	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes
Observations	5886	3781	3781	2687	3781
Adjusted R^2	0.299	0.137	0.517	0.337	0.559
F	18.629	13.114	0.937	1.804	0.867
Panel B: Low					
TreatYear × After	0.020 (0.026)	−0.033 (0.134)	0.048 (0.084)	−0.018 (0.251)	−0.019 (0.050)
TreatYear	0.008 (0.021)	−0.202 (0.134)	−0.028 (0.074)	−0.281 (0.219)	0.015 (0.044)
UserTenure	−0.037* (0.022)	−0.089 (0.094)	0.039 (0.050)	0.550*** (0.128)	0.021 (0.035)
UserAcceptedRate	0.009 (0.029)	−0.188 (0.154)	−0.064 (0.091)	0.254 (0.298)	−0.038 (0.060)
QuestionVolume	0.321*** (0.061)	0.247 (0.346)	−0.096 (0.159)	−0.455 (0.513)	−0.032 (0.101)
User FE	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes
Observations	3936	1566	1566	974	1566
Adjusted R^2	0.077	0.228	0.405	0.328	0.449
F	6.695	3.098	0.407	4.155	0.239

Notes: The table reports the results on answers created by users with different levels of past contributions. We count the answers created 20 days prior to the start of our analysis time window for each user. Standard errors are clustered at the user level

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

However, the consistently negative coefficients still provide some support for the existence of a decline in answer scores.

In summary, users who value their work and reputation appear to be the most affected, as the decline in engagement is more pronounced among high-reputation users compared to those whose status is based solely on seniority. These consistently stronger effects on active and reputable users provide supporting evidence that the decrease in answer volume and quality can be attributed to a decline in the motivation of loyal users to contribute answers.

5 Longer effect after the policy of banning ChatGPT

We intentionally focused on a short study period to avoid potential contamination from a platform-wide policy on December 5, 2022, which banned the use of ChatGPT. That said, we now investigate whether the decline in answer contributions, especially from heavy users, persisted even after the banning policy. This analysis helps strengthen our primary mechanism, the demotivating effect on loyal users.

The results shown in Fig. A5 and Table D2 show qualitatively similar findings when using an extended study period. After introducing an additional interaction term with a dummy variable for the period after the banning policy, we find that the significant negative effect on undeleted answer creation persists. For instance, column 4 of Table D2 indicates an approximately 4.8% drop in answer volume even in the 10 days following the implementation of the policy to remove suspected AI-generated answers. Nevertheless, our subgroup analysis of heavy users indicates that the negative effect decreased by approximately 25% following the implementation of the banning policy. The weakened effect suggests the policy's effectiveness in relieving users' concerns about contributing answers. However, the persistent negative impact may imply that loyal users still experience a decreasing sense of community belonging. Moreover, they might believe that others will refine the answers generated by ChatGPT more carefully to avoid removal, making it harder for them to distinguish their answers from others. Overall, this finding raises concerns for the platform because even after banning the AI tool, users, especially heavy and reputable users, do not contribute answers at the same level as before.

6 Concluding remarks

LLMs like ChatGPT have immense potential for generating text-based content and providing customized answers, leading to a critical shift in UGC platforms. The reduced costs of content creation facilitated by these models have increased user participation in content-based co-creation. However, this shift has also resulted in an influx of misinformation and low-quality content, leading to higher verification and moderation costs and discouraging the creation and consumption of high-quality content.

Using data from a programming Q&A platform, we examine the impact of generative AI on content creation. We observed a proliferation of AI-generated content on the platform following the introduction of ChatGPT. Importantly, we found that users

contributed fewer qualified and undeleted answers after ChatGPT's release. Beyond the displacement effect identified in prior research (Burtch et al., 2024), our analysis suggests the decline in question answers can also be attributed to a perceived increase in the sophistication of question presentation.

Further analyses indicate that the answer decline was primarily driven by highly engaged contributors who regularly supply solutions, signaling potential attrition among valued community participants. In contrast, infrequent users did not reduce their answer contributions. This divergence suggests differences in users' attitudes towards AI content generation. Loyal users appear cautious about the reliability of AI-generated content due to concerns about their reputation and the need to differentiate themselves, while inexperienced users actively embrace generative AI despite potential risks. Ultimately, those who have built their reputation through contributions appear most affected, as integrating AI-generated content not only threatens the expertise they have cultivated but also disrupts the whole community reputation system that determines their social rewards.

Our study has practical implications for individual content creators and online platforms. For individual creators, we found that generative AI tools have the potential to discourage heavy contributors, especially loyal users with strong community ties, from creating more content. As generative AI makes content creation easier, users may struggle to distinguish their own contributions from AI-generated material. This effect is likely more significant for dedicated users, whose creations could be overshadowed. In contrast, casual users may be more inclined to embrace these AI tools. To differentiate their content, creators should focus on building strong personal brands and fostering audience connections, in addition to producing high-quality work. Cultivating a distinctive identity and fostering deep bonds within the community can help ensure creators' contributions retain value, even as generative AI becomes more prevalent.

For online content platforms, our findings suggest a significant challenge—reduced content contributions, especially from heavy, loyal users. To address this, platforms could take several actions. First, platforms can enhance recognition for heavy users by offering micropayments or increased visibility of user status for activities like content refinement and answering follow-up inquiries. These incentives not only reward their efforts but also emphasize the unique value of their contributions different from AI content. Second, platforms could enforce AI-assisted content disclosure or automatically flag AI-generated content using detection algorithms. Overall, our study highlights the need for platforms to adapt their strategies to maintain high-quality, user-generated content in the face of increasingly accessible generative AI tools.

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Declarations

Conflict of Interest The authors declare no competing interests.

Ethics Approval Not applicable

Informed Consent Not applicable

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